



Fractions on Strips and Number Lines Teaching Bundle

Grade 3 • North Carolina • Mathematics

NC.3.NF.2 • Connect equal parts with fraction notation.

A focused classroom collection: task cards, five warm-ups, classwork, an exit ticket, four seasonal practice sets, substitute lessons, and planning pages. Each section has worked answers.

PRINT • TEACH • REVISIT



Your teaching route

Focus

Connect equal parts with fraction notation. This bundle practices a focused part of NC.3.NF.2; it is not a complete grade curriculum or a simulated EOG.

Before teaching

Learners should explain a simpler example using a model. Read the two worked examples, then use the first task card to check readiness.

Suggested sequence

Day 1: model for 10 minutes, then guided task cards for 15 minutes. Day 2: a 5-minute warm-up and 20 minutes of practice. Use the exit ticket to choose a small-group reteaching focus. Seasonal sets revisit the same skill using different quantities.

Materials

Pencils, blank paper, and graph paper or counters as useful. No paid app or internet is needed for student work. A calculator may check finished calculations; these are instructional choices, not official testing accommodation rules.



Make the reasoning visible

Choose a whole and partition it equally. The denominator counts all equal parts; the numerator counts selected parts.

Example 1: A strip is divided into 2 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

Each part is $\frac{1}{2}$. 2 copies make $\frac{2}{2}$. The point is 2 equal steps of size $\frac{1}{2}$ from zero.

Example 2: A strip is divided into 2 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

Each part is $\frac{1}{2}$. 1 copy makes $\frac{1}{2}$. The point is 1 equal step of size $\frac{1}{2}$ from zero.

Ask: What does each number represent? How could a model or inverse operation check the result?

Task cards 1–4

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 01

A strip is divided into 4 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

CARD 02

A strip is divided into 8 equal parts. 6 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

CARD 03

A strip is divided into 6 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

CARD 04

A strip is divided into 2 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

Task cards 5–8

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 05

A strip is divided into 2 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

CARD 06

A strip is divided into 8 equal parts. 8 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

CARD 07

A strip is divided into 3 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

CARD 08

A strip is divided into 2 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

Task cards 9–12

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 09

A strip is divided into 6 equal parts. 6 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

CARD 10

A strip is divided into 3 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

CARD 11

A strip is divided into 4 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

CARD 12

A strip is divided into 3 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

Task cards 13–16

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 13

A strip is divided into 6 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

CARD 14

A strip is divided into 6 equal parts. 4 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

CARD 15

A strip is divided into 4 equal parts. 4 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

CARD 16

A strip is divided into 8 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Explain your method.

Task-card response record

Use this page for final answers; attach drawings or full worked solutions on separate paper.

1.

2.

3.

4.

5.

6.

7.

8.

9.

10.

11.

12.

13.

14.

15.

16.



Warm-up • Day 1

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

1. A strip is divided into 4 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



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2. A strip is divided into 6 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Warm-up • Day 2

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

3. A strip is divided into 3 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



4. A strip is divided into 4 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Warm-up • Day 3

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

5. A strip is divided into 4 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



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6. A strip is divided into 4 equal parts. 4 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Warm-up • Day 4

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

7. A strip is divided into 6 equal parts. 4 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



8. A strip is divided into 3 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Warm-up • Day 5

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

9. A strip is divided into 8 equal parts. 5 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



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10. A strip is divided into 4 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A strip is divided into 3 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



2. A strip is divided into 8 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



3. A strip is divided into 8 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. A strip is divided into 6 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



5. A strip is divided into 4 equal parts. 4 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



6. A strip is divided into 3 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Exit ticket • Try independently

Complete the four questions independently. Show enough reasoning for your teacher to see your method. This is a classroom check, not a standardized score.

1. A strip is divided into 4 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

2. A strip is divided into 8 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

3. A strip is divided into 6 equal parts. 5 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

4. A strip is divided into 8 equal parts. 8 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Task-card worked solutions

1. $\frac{1}{4}$

Each part is $\frac{1}{4}$. 1 copy makes $\frac{1}{4}$. The point is 1 equal step of size $\frac{1}{4}$ from zero.

2. $\frac{6}{8}$

Each part is $\frac{1}{8}$. 6 copies make $\frac{6}{8}$. The point is 6 equal steps of size $\frac{1}{8}$ from zero.

3. $\frac{2}{6}$

Each part is $\frac{1}{6}$. 2 copies make $\frac{2}{6}$. The point is 2 equal steps of size $\frac{1}{6}$ from zero.

4. $\frac{2}{2}$

Each part is $\frac{1}{2}$. 2 copies make $\frac{2}{2}$. The point is 2 equal steps of size $\frac{1}{2}$ from zero.

5. $\frac{2}{2}$

Each part is $\frac{1}{2}$. 2 copies make $\frac{2}{2}$. The point is 2 equal steps of size $\frac{1}{2}$ from zero.

6. $\frac{8}{8}$

Each part is $\frac{1}{8}$. 8 copies make $\frac{8}{8}$. The point is 8 equal steps of size $\frac{1}{8}$ from zero.

Task-card worked solutions

7. $\frac{1}{3}$

Each part is $\frac{1}{3}$. 1 copy makes $\frac{1}{3}$. The point is 1 equal step of size $\frac{1}{3}$ from zero.

8. $\frac{2}{2}$

Each part is $\frac{1}{2}$. 2 copies make $\frac{2}{2}$. The point is 2 equal steps of size $\frac{1}{2}$ from zero.

9. $\frac{6}{6}$

Each part is $\frac{1}{6}$. 6 copies make $\frac{6}{6}$. The point is 6 equal steps of size $\frac{1}{6}$ from zero.

10. $\frac{3}{3}$

Each part is $\frac{1}{3}$. 3 copies make $\frac{3}{3}$. The point is 3 equal steps of size $\frac{1}{3}$ from zero.

11. $\frac{3}{4}$

Each part is $\frac{1}{4}$. 3 copies make $\frac{3}{4}$. The point is 3 equal steps of size $\frac{1}{4}$ from zero.

12. $\frac{2}{3}$

Each part is $\frac{1}{3}$. 2 copies make $\frac{2}{3}$. The point is 2 equal steps of size $\frac{1}{3}$ from zero.

Task-card worked solutions

13. $\frac{2}{6}$

Each part is $\frac{1}{6}$. 2 copies make $\frac{2}{6}$. The point is 2 equal steps of size $\frac{1}{6}$ from zero.

14. $\frac{4}{6}$

Each part is $\frac{1}{6}$. 4 copies make $\frac{4}{6}$. The point is 4 equal steps of size $\frac{1}{6}$ from zero.

15. $\frac{4}{4}$

Each part is $\frac{1}{4}$. 4 copies make $\frac{4}{4}$. The point is 4 equal steps of size $\frac{1}{4}$ from zero.

16. $\frac{2}{8}$

Each part is $\frac{1}{8}$. 2 copies make $\frac{2}{8}$. The point is 2 equal steps of size $\frac{1}{8}$ from zero.

Warm-up worked solutions

1. $\frac{1}{4}$

Each part is $\frac{1}{4}$. 1 copy makes $\frac{1}{4}$. The point is 1 equal step of size $\frac{1}{4}$ from zero.

2. $\frac{1}{6}$

Each part is $\frac{1}{6}$. 1 copy makes $\frac{1}{6}$. The point is 1 equal step of size $\frac{1}{6}$ from zero.

3. $\frac{3}{3}$

Each part is $\frac{1}{3}$. 3 copies make $\frac{3}{3}$. The point is 3 equal steps of size $\frac{1}{3}$ from zero.

4. $\frac{2}{4}$

Each part is $\frac{1}{4}$. 2 copies make $\frac{2}{4}$. The point is 2 equal steps of size $\frac{1}{4}$ from zero.

5. $\frac{1}{4}$

Each part is $\frac{1}{4}$. 1 copy makes $\frac{1}{4}$. The point is 1 equal step of size $\frac{1}{4}$ from zero.

6. $\frac{4}{4}$

Each part is $\frac{1}{4}$. 4 copies make $\frac{4}{4}$. The point is 4 equal steps of size $\frac{1}{4}$ from zero.

Warm-up worked solutions

7. $\frac{4}{6}$

Each part is $\frac{1}{6}$. 4 copies make $\frac{4}{6}$. The point is 4 equal steps of size $\frac{1}{6}$ from zero.

8. $\frac{3}{3}$

Each part is $\frac{1}{3}$. 3 copies make $\frac{3}{3}$. The point is 3 equal steps of size $\frac{1}{3}$ from zero.

9. $\frac{5}{8}$

Each part is $\frac{1}{8}$. 5 copies make $\frac{5}{8}$. The point is 5 equal steps of size $\frac{1}{8}$ from zero.

10. $\frac{3}{4}$

Each part is $\frac{1}{4}$. 3 copies make $\frac{3}{4}$. The point is 3 equal steps of size $\frac{1}{4}$ from zero.

Classwork worked solutions

1. $\frac{3}{3}$

Each part is $\frac{1}{3}$. 3 copies make $\frac{3}{3}$. The point is 3 equal steps of size $\frac{1}{3}$ from zero.

2. $\frac{3}{8}$

Each part is $\frac{1}{8}$. 3 copies make $\frac{3}{8}$. The point is 3 equal steps of size $\frac{1}{8}$ from zero.

3. $\frac{3}{8}$

Each part is $\frac{1}{8}$. 3 copies make $\frac{3}{8}$. The point is 3 equal steps of size $\frac{1}{8}$ from zero.

4. $\frac{1}{6}$

Each part is $\frac{1}{6}$. 1 copy makes $\frac{1}{6}$. The point is 1 equal step of size $\frac{1}{6}$ from zero.

5. $\frac{4}{4}$

Each part is $\frac{1}{4}$. 4 copies make $\frac{4}{4}$. The point is 4 equal steps of size $\frac{1}{4}$ from zero.

6. $\frac{3}{3}$

Each part is $\frac{1}{3}$. 3 copies make $\frac{3}{3}$. The point is 3 equal steps of size $\frac{1}{3}$ from zero.

Exit-ticket worked solutions

1. $\frac{2}{4}$

Each part is $\frac{1}{4}$. 2 copies make $\frac{2}{4}$. The point is 2 equal steps of size $\frac{1}{4}$ from zero.

2. $\frac{1}{8}$

Each part is $\frac{1}{8}$. 1 copy makes $\frac{1}{8}$. The point is 1 equal step of size $\frac{1}{8}$ from zero.

3. $\frac{5}{6}$

Each part is $\frac{1}{6}$. 5 copies make $\frac{5}{6}$. The point is 5 equal steps of size $\frac{1}{6}$ from zero.

4. $\frac{8}{8}$

Each part is $\frac{1}{8}$. 8 copies make $\frac{8}{8}$. The point is 8 equal steps of size $\frac{1}{8}$ from zero.

Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A strip is divided into 8 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

2. A strip is divided into 2 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

3. A strip is divided into 2 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

4. A strip is divided into 6 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A strip is divided into 2 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

6. A strip is divided into 8 equal parts. 7 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

7. A strip is divided into 4 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

8. A strip is divided into 4 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Fall • Worked solutions

1. $\frac{2}{8}$

Each part is $\frac{1}{8}$. 2 copies make $\frac{2}{8}$. The point is 2 equal steps of size $\frac{1}{8}$ from zero.

2. $\frac{1}{2}$

Each part is $\frac{1}{2}$. 1 copy makes $\frac{1}{2}$. The point is 1 equal step of size $\frac{1}{2}$ from zero.

3. $\frac{2}{2}$

Each part is $\frac{1}{2}$. 2 copies make $\frac{2}{2}$. The point is 2 equal steps of size $\frac{1}{2}$ from zero.

4. $\frac{2}{6}$

Each part is $\frac{1}{6}$. 2 copies make $\frac{2}{6}$. The point is 2 equal steps of size $\frac{1}{6}$ from zero.



Fall • Worked solutions

5. $\frac{2}{2}$

Each part is $\frac{1}{2}$. 2 copies make $\frac{2}{2}$. The point is 2 equal steps of size $\frac{1}{2}$ from zero.

6. $\frac{7}{8}$

Each part is $\frac{1}{8}$. 7 copies make $\frac{7}{8}$. The point is 7 equal steps of size $\frac{1}{8}$ from zero.

7. $\frac{3}{4}$

Each part is $\frac{1}{4}$. 3 copies make $\frac{3}{4}$. The point is 3 equal steps of size $\frac{1}{4}$ from zero.

8. $\frac{3}{4}$

Each part is $\frac{1}{4}$. 3 copies make $\frac{3}{4}$. The point is 3 equal steps of size $\frac{1}{4}$ from zero.

Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A strip is divided into 2 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

2. A strip is divided into 2 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

3. A strip is divided into 8 equal parts. 6 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

4. A strip is divided into 3 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A strip is divided into 2 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

6. A strip is divided into 6 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

7. A strip is divided into 8 equal parts. 6 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

8. A strip is divided into 8 equal parts. 4 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Winter • Worked solutions

1. $\frac{2}{2}$

Each part is $\frac{1}{2}$. 2 copies make $\frac{2}{2}$. The point is 2 equal steps of size $\frac{1}{2}$ from zero.

2. $\frac{1}{2}$

Each part is $\frac{1}{2}$. 1 copy makes $\frac{1}{2}$. The point is 1 equal step of size $\frac{1}{2}$ from zero.

3. $\frac{6}{8}$

Each part is $\frac{1}{8}$. 6 copies make $\frac{6}{8}$. The point is 6 equal steps of size $\frac{1}{8}$ from zero.

4. $\frac{2}{3}$

Each part is $\frac{1}{3}$. 2 copies make $\frac{2}{3}$. The point is 2 equal steps of size $\frac{1}{3}$ from zero.

Winter • Worked solutions

5. $\frac{1}{2}$

Each part is $\frac{1}{2}$. 1 copy makes $\frac{1}{2}$. The point is 1 equal step of size $\frac{1}{2}$ from zero.

6. $\frac{3}{6}$

Each part is $\frac{1}{6}$. 3 copies make $\frac{3}{6}$. The point is 3 equal steps of size $\frac{1}{6}$ from zero.

7. $\frac{6}{8}$

Each part is $\frac{1}{8}$. 6 copies make $\frac{6}{8}$. The point is 6 equal steps of size $\frac{1}{8}$ from zero.

8. $\frac{4}{8}$

Each part is $\frac{1}{8}$. 4 copies make $\frac{4}{8}$. The point is 4 equal steps of size $\frac{1}{8}$ from zero.

Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A strip is divided into 4 equal parts. 4 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

2. A strip is divided into 4 equal parts. 4 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

3. A strip is divided into 6 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

4. A strip is divided into 4 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A strip is divided into 2 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

6. A strip is divided into 3 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

7. A strip is divided into 8 equal parts. 7 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

8. A strip is divided into 4 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Spring break • Worked solutions

1. $\frac{4}{4}$

Each part is $\frac{1}{4}$. 4 copies make $\frac{4}{4}$. The point is 4 equal steps of size $\frac{1}{4}$ from zero.

2. $\frac{4}{4}$

Each part is $\frac{1}{4}$. 4 copies make $\frac{4}{4}$. The point is 4 equal steps of size $\frac{1}{4}$ from zero.

3. $\frac{2}{6}$

Each part is $\frac{1}{6}$. 2 copies make $\frac{2}{6}$. The point is 2 equal steps of size $\frac{1}{6}$ from zero.

4. $\frac{2}{4}$

Each part is $\frac{1}{4}$. 2 copies make $\frac{2}{4}$. The point is 2 equal steps of size $\frac{1}{4}$ from zero.



Spring break • Worked solutions

5. $\frac{2}{2}$

Each part is $\frac{1}{2}$. 2 copies make $\frac{2}{2}$. The point is 2 equal steps of size $\frac{1}{2}$ from zero.

6. $\frac{3}{3}$

Each part is $\frac{1}{3}$. 3 copies make $\frac{3}{3}$. The point is 3 equal steps of size $\frac{1}{3}$ from zero.

7. $\frac{7}{8}$

Each part is $\frac{1}{8}$. 7 copies make $\frac{7}{8}$. The point is 7 equal steps of size $\frac{1}{8}$ from zero.

8. $\frac{1}{4}$

Each part is $\frac{1}{4}$. 1 copy makes $\frac{1}{4}$. The point is 1 equal step of size $\frac{1}{4}$ from zero.

Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A strip is divided into 3 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

2. A strip is divided into 4 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

3. A strip is divided into 4 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

4. A strip is divided into 4 equal parts. 4 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A strip is divided into 6 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

6. A strip is divided into 3 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

7. A strip is divided into 3 equal parts. 3 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

8. A strip is divided into 8 equal parts. 6 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Summer • Worked solutions

1. $\frac{3}{3}$

Each part is $\frac{1}{3}$. 3 copies make $\frac{3}{3}$. The point is 3 equal steps of size $\frac{1}{3}$ from zero.

2. $\frac{1}{4}$

Each part is $\frac{1}{4}$. 1 copy makes $\frac{1}{4}$. The point is 1 equal step of size $\frac{1}{4}$ from zero.

3. $\frac{3}{4}$

Each part is $\frac{1}{4}$. 3 copies make $\frac{3}{4}$. The point is 3 equal steps of size $\frac{1}{4}$ from zero.

4. $\frac{4}{4}$

Each part is $\frac{1}{4}$. 4 copies make $\frac{4}{4}$. The point is 4 equal steps of size $\frac{1}{4}$ from zero.

Summer • Worked solutions

5. $\frac{2}{6}$

Each part is $\frac{1}{6}$. 2 copies make $\frac{2}{6}$. The point is 2 equal steps of size $\frac{1}{6}$ from zero.

6. $\frac{2}{3}$

Each part is $\frac{1}{3}$. 2 copies make $\frac{2}{3}$. The point is 2 equal steps of size $\frac{1}{3}$ from zero.

7. $\frac{3}{3}$

Each part is $\frac{1}{3}$. 3 copies make $\frac{3}{3}$. The point is 3 equal steps of size $\frac{1}{3}$ from zero.

8. $\frac{6}{8}$

Each part is $\frac{1}{8}$. 6 copies make $\frac{6}{8}$. The point is 6 equal steps of size $\frac{1}{8}$ from zero.

Substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Connect equal parts with fraction notation. Let students explain a familiar strategy to a partner.

5-15 minutes: model

Use the worked-example page. Choose a whole and partition it equally. The denominator counts all equal parts; the numerator counts selected parts.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A strip is divided into 8 equal parts. 5 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

2. A strip is divided into 3 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

3. A strip is divided into 6 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

4. A strip is divided into 2 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Substitute lesson • Worked solutions

1. $\frac{5}{8}$

Each part is $\frac{1}{8}$. 5 copies make $\frac{5}{8}$. The point is 5 equal steps of size $\frac{1}{8}$ from zero.

2. $\frac{2}{3}$

Each part is $\frac{1}{3}$. 2 copies make $\frac{2}{3}$. The point is 2 equal steps of size $\frac{1}{3}$ from zero.

3. $\frac{2}{6}$

Each part is $\frac{1}{6}$. 2 copies make $\frac{2}{6}$. The point is 2 equal steps of size $\frac{1}{6}$ from zero.

4. $\frac{2}{2}$

Each part is $\frac{1}{2}$. 2 copies make $\frac{2}{2}$. The point is 2 equal steps of size $\frac{1}{2}$ from zero.

Emergency substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Connect equal parts with fraction notation. Let students explain a familiar strategy to a partner.

5-15 minutes: model

No copies or devices? Write the first worked example on the board; students use blank paper. Choose a whole and partition it equally. The denominator counts all equal parts; the numerator counts selected parts.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Emergency substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A strip is divided into 8 equal parts. 8 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

2. A strip is divided into 8 equal parts. 7 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

3. A strip is divided into 8 equal parts. 6 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

4. A strip is divided into 3 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.



Emergency substitute lesson • Worked solutions

1. $\frac{8}{8}$

Each part is $\frac{1}{8}$. 8 copies make $\frac{8}{8}$. The point is 8 equal steps of size $\frac{1}{8}$ from zero.

2. $\frac{7}{8}$

Each part is $\frac{1}{8}$. 7 copies make $\frac{7}{8}$. The point is 7 equal steps of size $\frac{1}{8}$ from zero.

3. $\frac{6}{8}$

Each part is $\frac{1}{8}$. 6 copies make $\frac{6}{8}$. The point is 6 equal steps of size $\frac{1}{8}$ from zero.

4. $\frac{1}{3}$

Each part is $\frac{1}{3}$. 1 copy makes $\frac{1}{3}$. The point is 1 equal step of size $\frac{1}{3}$ from zero.

A completed lesson-plan example

Level and focus

Grade 3; NC.3.NF.2; Connect equal parts with fraction notation.

Evidence of learning

Student represents the quantities, selects a valid method, calculates accurately, and explains the result with appropriate units.

Launch / model

Use worked example 1. Ask students to annotate the role of every quantity. Demonstrate the model before moving to symbols.

Guided / independent practice

Use cards 1-4 in pairs; assign classwork independently. Change grouping according to the explanation you hear rather than speed alone.

Formative check

Use the four-question exit ticket. Score each item 0-2: 2 for an accurate result with coherent reasoning, 1 for a sound method with a minor error or an unexplained correct result, 0 for missing or unsupported reasoning.

Next teaching decision

0-3 points: reteach with a concrete model. 4-6: correct the specific misconception with a short example. 7-8: invite a second representation or a student-created problem. These are local instructional decisions, not mastery cutoffs validated by a test publisher.

Plan your next lesson

Date, class, and focus / exact local standard:

What will students do to show understanding?

Launch and worked example:

Guided practice and independent task:

Access supports and extension:

Exit evidence and next lesson decision:



Access, sources, and classroom license

AIG: Invite two methods, a counterexample, or a new task with a specified solution. Ask students to explain limits of a shortcut.

504: Apply the student's plan: breaks, extra time, enlarged print, reduced copying, or alternate response methods as documented.

EC: Use the IEP to select supports. Model one step, offer counters or a diagram, reduce the visible item set, and accept an oral explanation when appropriate.

ESL: Preview quantity, difference, factor, or other relevant terms. Let students rehearse in a home language, then use "I chose ___ because ___." Maintain the mathematical goal.

Official alignment sources • accessed September 15, 2026

NC.3.NF.2 is verified in the NCDPI mathematics progression. The activity objective is original wording; it describes the skill focus, not the full standard text.

NCDPI mathematics standards:

<https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>

NCDPI EOG mathematics specifications:

<https://www.dpi.nc.gov/documents/accountability/testing/eog/eog-mathematics-grades-3-8-test-specifications>

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